D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

Discuss the role of HIV in the pathogenesis of Cancer. Discuss the cancers associated with HIV infection. Discuss in detail the management of various lymphomas arising in a patient with HIV-AIDS.

August 2022 | DM 0822 | Q1 (DEEP Essay / Elaborate (FOCUS 2022–2026)) | **15 marks** | Target ~16 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

0. How to use this DEEP answer

Sections below are **high-yield, trial-dense content** written for a 15-mark Paper IV elaboration. After the deep core, additional revision modules (algorithms, supportive care, flash cards) extend the note to full booklet length.

EXAM PEARLS / HALLMARKS

- In the exam hall, prioritise the DEEP trial tables + NCCN 2026 boxes first.
- Draw at least one flowchart from the deep sections.

1. Introduction — HIV, immunodeficiency & cancer

HIV-1 infection causes progressive CD4+ T-cell depletion, chronic immune activation, and loss of immune surveillance. Cancer risk rises from oncogenic co-infections, impaired EBV/HHV8/HPV control, and lifestyle confounders (tobacco). ART has shifted the spectrum from AIDS-defining cancers toward non-AIDS-defining cancers (NADCs) as people live longer.

Essay structure demanded by the question: (A) pathogenesis, (B) cancer list AIDS-defining vs NADC, (C) detailed HIV lymphoma management (DLBCL, BL, PEL, PBL, HL).

- AIDS-defining: Kaposi sarcoma, non-Hodgkin lymphoma, invasive cervical cancer.
- Always start/continue ART — cornerstone of both prevention and lymphoma outcomes.

EXAM PEARLS / HALLMARKS

- Opening line: ‘HIV is both an oncogenic facilitator and a treatable comorbidity — ART plus standard lymphoma regimens.’

2. Pathogenesis — mechanisms linking HIV to cancer

Mechanisms: (1) loss of virus-specific CD4/CD8 control of EBV and HHV8; (2) chronic B-cell activation and dysregulation → lymphomagenesis; (3) HIV proteins (Tat, Nef) contributing to angiogenesis/inflammation in KS pathogenesis alongside HHV8; (4) mucosal HPV persistence → cervical/anal dysplasia; (5) HBV/HCV co-infection → HCC; (6) immune reconstitution later changing KS/lymphoma presentations.

Direct HIV integration into oncogenes is not the dominant solid-tumor mechanism; **indirect immunodeficiency + co-infection** predominates.

- Nadir CD4 and uncontrolled viremia correlate with lymphoma/KS risk.
- ART reduces AIDS-defining cancers dramatically but NADCs persist.

CD4 depletion	Failed immune surveillance
Chronic inflammation	Activation-induced cytidine deaminase / mutagenesis milieu
EBV co-infection	BL, many DLBCL, HL biology
HHV8/KSHV	KS, PEL, some multicentric Castleman
HPV / HBV / HCV	Cervical, anal, HCC

Figure: mechanism / pathway map

3. Cancers associated with HIV — spectrum

AIDS-defining: Kaposi sarcoma; NHL (including BL, DLBCL, PEL, primary CNS lymphoma); invasive cervical carcinoma. **NADCs with elevated SIR**: Hodgkin lymphoma, anal cancer, lung cancer, HCC, head-and-neck (HPV), skin cancers; relative risks vary by region and smoking prevalence.

Indian context: suspect HIV in unusual KS, PEL, PBL, primary CNS lymphoma, or aggressive extranodal NHL in young adults; offer testing broadly per NACO counseling norms.

- Primary CNS lymphoma in HIV: usually EBV+, CD4 often <50 — distinguish from toxoplasmosis.
- PBL: oral/jaw masses in low CD4; MYC; plasmablastic phenotype (CD20 often negative).
- PEL: body-cavity effusions; HHV8+; often EBV co-positive; CD20 variable/negative.

HIV testing and parallel ART/oncology care

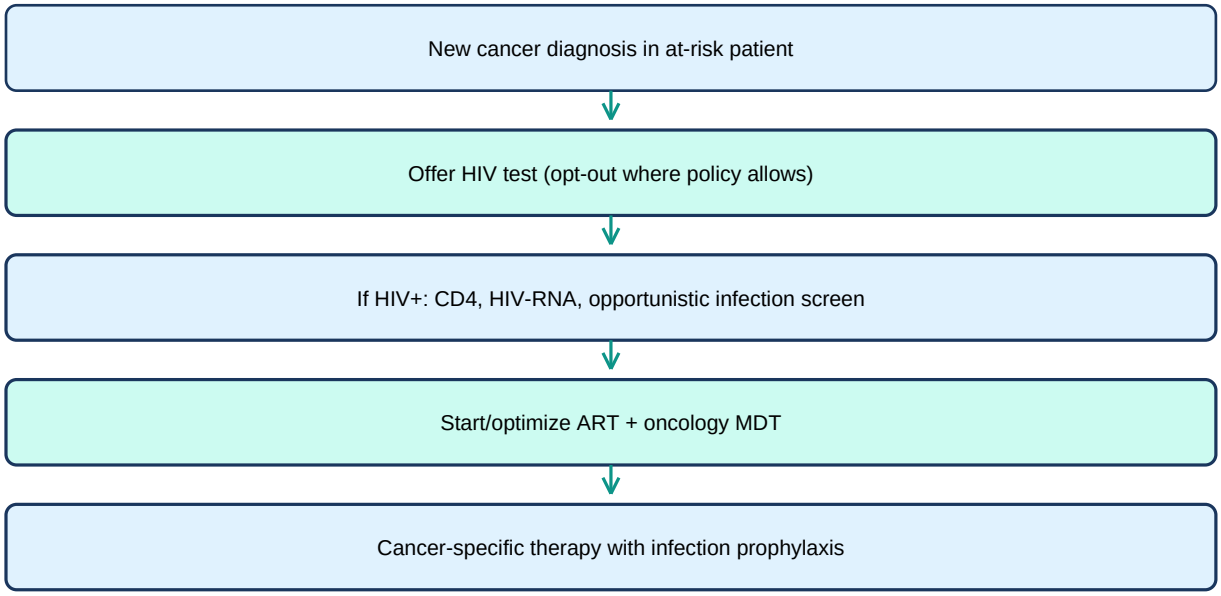


Figure: stepwise exam algorithm

4. General principles of lymphoma care in HIV-AIDS

Modern paradigm: treat HIV-associated lymphomas with **full-dose standard regimens** plus concurrent ART, growth-factor support, and opportunistic infection prophylaxis — outcomes approach non-HIV patients when CD4 and performance status allow.

Baseline: CBC, chemistries, LDH, uric acid, viral hepatitis, CMV IgG, Toxoplasma IgG, β2-microglobulin, CT/PET-CT, bone marrow as indicated, ECG/echo, pregnancy test, fertility counseling.

- PJP prophylaxis (TMP-SMX) for all on intensive chemo or CD4 <200.

• Azole/antiviral choices: watch ART–chemo CYP interactions (especially vinca, midazolam, some TKIs).

• Rituximab: use if CD20+; caution historically if CD4 very low — modern practice still favors R if CD20+ with infection vigilance (AMC data nuance).

Drug / regimen	Dose / schedule (exam pearls)
TMP-SMX prophylaxis	1 DS tablet thrice weekly or daily per protocol
Filgrastim support	As needed to deliver on-time cycles
Intrathecal MTX/cytarabine	CNS prophylaxis per histology risk (BL, extranodal)

Table: doses and regimens

5. Diffuse large B-cell lymphoma (HIV-DLBCL)

Most common HIV NHL. Often extranodal, high IPI, EBV±. Regimens: **R-CHOP** or **DA-EPOCH-R** (NCI infusional regimen widely used in HIV). AMC and NCI experience support EPOCH-based therapy with concurrent ART and rituximab in CD20+ disease.

CNS prophylaxis for high-risk features (extranodal sites, elevated LDH, MYC rearrangements). Restage with PET-CT; consolidate with ASCT in chemosensitive relapse if ART-controlled and stem-cell feasible.

• Do not empirically dose-reduce solely because of HIV label — fitness and organ function decide.

• If CD20-negative plasmablastic morphology → see PBL section (no rituximab benefit).

Trial	Key result / exam takeaway
NCI DA-EPOCH-R HIV cohorts	High CR rates with concurrent ART
AMC rituximab studies	R improves responses in CD20+ HIV-DLBCL with infection monitoring

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
R-CHOP	Rituximab 375 mg/m² + CHOP q21d ×6 (HIV: G-CSF liberal)
DA-EPOCH-R	Infusional EPOCH + rituximab; dose-adjust to ANC/platelets nadir
IT chemotherapy	Per protocol days for CNS risk

Table: doses and regimens

6. Burkitt lymphoma in HIV

Highly aggressive; MYC translocation; often EBV+. Treat with intensive regimens: CODOX-M/IVAC ± R, or DA-EPOCH-R, with mandatory CNS prophylaxis. TLS risk extreme — hydrate, rasburicase if high-risk, telemetry labs.

Outcomes excellent if treated intensively with ART; avoid CHOP alone for BL.

HIV-Burkitt management skeleton

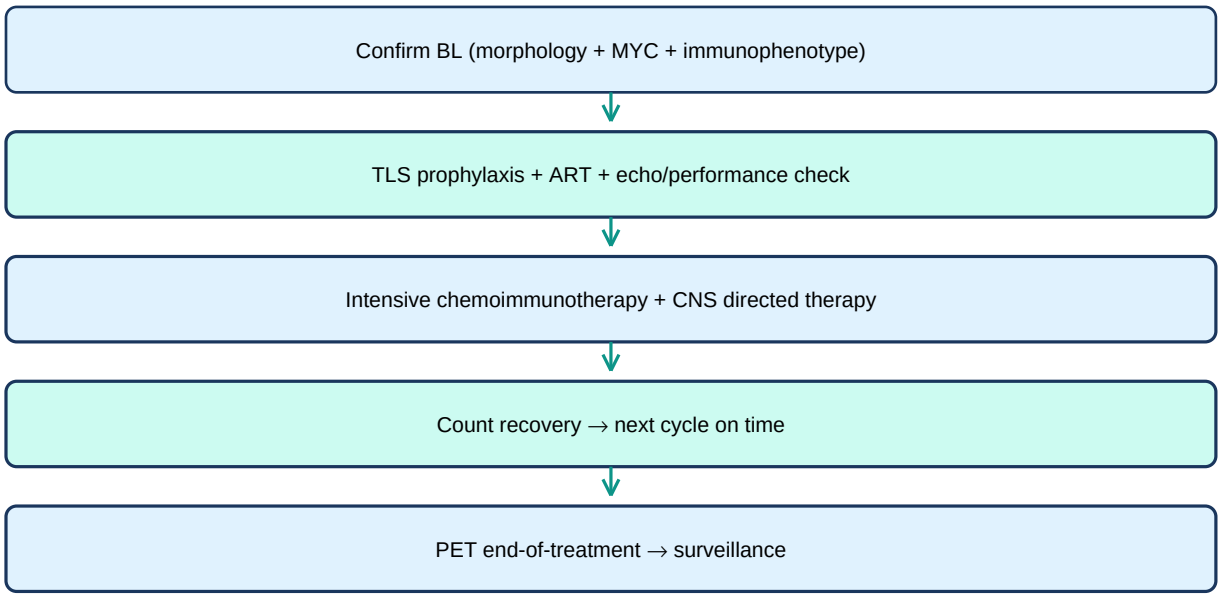


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

• Exam red flag: CHOP for BL = inadequate.

7. Primary effusion lymphoma (PEL)

HHV8-positive effusion-based lymphoma (pleural/peritoneal/pericardial) without solid masses usually; extracavitary PEL variants exist. Immunophenotype often null for B-cell markers (CD20–), CD30±, CD138±, EMA±. EBV frequently co-positive.

Therapy: ART essential; CHOP-like or EPOCH-based regimens used; prognosis historically poor. Rituximab generally not useful if CD20–. Consider extracavitary disease radiotherapy in selected cases; clinical trials preferred.

- Send HHV8 LANA IHC / PCR on cytology cell block.
- Differentiate from HHV8+ multicentric Castleman disease (MCD) — different treatment (rituximab ± antivirals/ART).

HHV8 latent proteins	Drive PEL survival programs
Effusion microenvironment	IL-6 rich, poor drug penetration challenges
ART	Immune recovery + HHV8 control adjunct

Figure: mechanism / pathway map

8. Plasmablastic lymphoma (PBL)

Aggressive CD20-negative lymphoma with plasmablastic morphology; frequent oral cavity involvement; MYC rearrangements common; EBV often +. Associated with low CD4.

Treatment: intensive regimens (EPOCH, CODOX-based, or myeloma-like approaches in selected reports) + ART; rituximab usually omitted. Prognosis guarded; CNS prophylaxis individualized.

- Do not mistake for myeloma — clinical context + HIV + EBV help.
- Supportive care intensive: mucositis, infections, TLS.

Trial	Key result / exam takeaway
Retrospective EPOCH series in PBL	Best available evidence; no large RCT

Table: landmark clinical trials

9. Primary CNS lymphoma (HIV-PCNSL)

Typically EBV-associated DLBCL confined to CNS; CD4 often very low. Differentiate from toxoplasma abscess (multiple ring-enhancing lesions, Toxo IgG+). CSF EBV PCR supportive; biopsy when safe.

Treat with high-dose methotrexate-based regimens ± rituximab as tolerated, concurrent ART, and infection prophylaxis. WBRT historically used but neurotoxicity high — prefer chemo-first when feasible.

- Empiric Toxoplasma therapy sometimes started pending diagnosis in unstable patients — center-dependent.
- ART can produce IRIS — monitor neurologic status.

10. Hodgkin lymphoma in HIV

HL is a NADC with markedly increased incidence; often mixed-cellularity, advanced stage, EBV+. ABVD remains a standard backbone with ART and G-CSF as needed; brentuximab vedotin combinations and PD-1 inhibitors enter R/R settings with careful infection/IRIS awareness.

PET-adapted strategies as in non-HIV HL when logistics allow; avoid bleomycin lung toxicity with growth-factor caution (bleomycin + G-CSF interaction debated — follow protocol).

- Outcomes approach HIV-negative HL with modern ART.
- Screen for TB before IO if endemic risk.

Drug / regimen	Dose / schedule (exam pearls)
ABVD	Standard doses q14d × up to 6 cycles with ART
Brentuximab vedotin	1.8 mg/kg q3w in BV-based regimens / R/R
Nivolumab / pembrolizumab	R/R HL — monitor irAE + HIV control

Table: doses and regimens

11. Kaposi sarcoma & MCD (brief, for completeness)

KS: HHV8-driven; ART is first-line for limited disease; liposomal doxorubicin or paclitaxel for advanced. Pomalidomide is a modern systemic option in some labels/guidelines. HHV8-MCD: rituximab-based therapy + ART; watch for concurrent lymphoma.

- Pulmonary KS: treat promptly — high morbidity.
- IRIS-KS after ART initiation — continue ART, treat KS.

12. ART interactions & infection prophylaxis checklist

Prefer ART regimens with fewer CYP3A interactions when combining with vincristine/vinblastine (avoid boosting conflicts when possible — involve HIV pharmacist). Integrase inhibitor–based ART commonly used. Never interrupt ART casually during chemo.

- PJP prophylaxis; MAC prophylaxis if CD4 <50 (azithromycin) per guidelines.
- HBV co-infection: entecavir/tenofovir backbone to prevent reactivation.
- Fungal prophylaxis individualized for prolonged neutropenia.

ART–chemo safety pathway

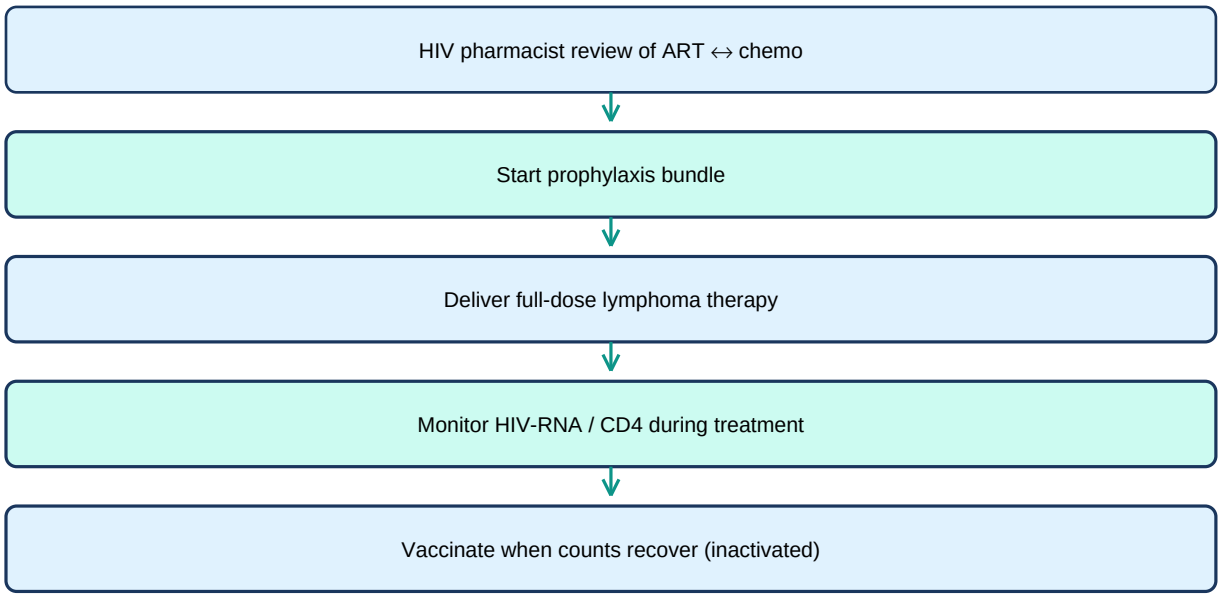


Figure: stepwise exam algorithm

13. Relapsed disease, transplant & cellular therapy

Salvage chemo + ASCT is appropriate for chemosensitive relapsed HIV-DLBCL/HL with controlled HIV at experienced centers. CAR-T and bispecifics: emerging case series in HIV+ lymphoma if viral control adequate — discuss trials; not yet universal standard in all Indian centers.

- Exclude opportunistic infection before intensive salvage.
- Donor/registry issues historically limited allo-SCT; selected cases possible.

14. NCCN / guideline bullets (lymphoma & HIV practice)

Cite NCCN B-Cell Lymphomas / Hodgkin guidelines for regimen selection and note HIV-specific principles: concurrent ART, prophylaxis, and preference for standard intensive therapy. NCCN 2026 B-cell updates (bispecifics, CAR-T sequencing) apply to R/R pathways when HIV controlled.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* B-Cell Lymphomas v4.2026; Hodgkin Lymphoma v2.2026
- Treat HIV-associated B-cell lymphomas with standard NCCN regimens plus ART rather than attenuated ‘AIDS-era’ palliative chemo when fitness allows.
- R/R B-cell: consider bispecifics/CAR-T pathways per NCCN 2026 when accessible and HIV controlled.
- HL: ABVD/BV-based and PD-1 pathways as indicated; integrate infection risk assessment.
- Document CD4/HIV-RNA alongside Lugano response assessments.

15. Indian / NACO / NCG practical notes

Coordinate with NACO ART centers for uninterrupted antiretrovirals. Generic chemo and biosimilar rituximab improve access. TB co-infection endemic — screen before steroids/IO. PCR for EBV/HHV8 and PET may be limited peripherally — refer tertiary centers for PEL/PBL/PCNSL.

- Offer HIV testing in aggressive extranodal NHL and oral PBL presentations.
- Counsel stigma-sensitive care; involve counselors.

16. Summary & exam pearls

HIV promotes cancer via immunodeficiency and oncogenic viruses. AIDS-defining cancers and NADCs both matter. Lymphoma care = ART + full-dose histology-specific therapy (R-CHOP/DA-EPOCH-R, intensive BL regimens, specialized PEL/PBL/PCNSL approaches, ABVD for HL) with rigorous infection prophylaxis.

EXAM PEARLS / HALLMARKS

- ART concurrent — non-negotiable.
- BL ≠ CHOP; use intensive regimens + CNS prophylaxis.
- PEL = HHV8 effusion lymphoma, often CD20–.
- PBL = oral, CD20–, MYC, EBV — no rituximab.
- PCNSL: HD-MTX ± R + ART; rule out Toxo.
- Rituximab if CD20+ with infection vigilance.

— Additional exam-length revision modules —

The following modules reinforce algorithms, toxicities, Indian/NCG context, and last-page summary structure.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

Work-up algorithm (draw this in the answer booklet)

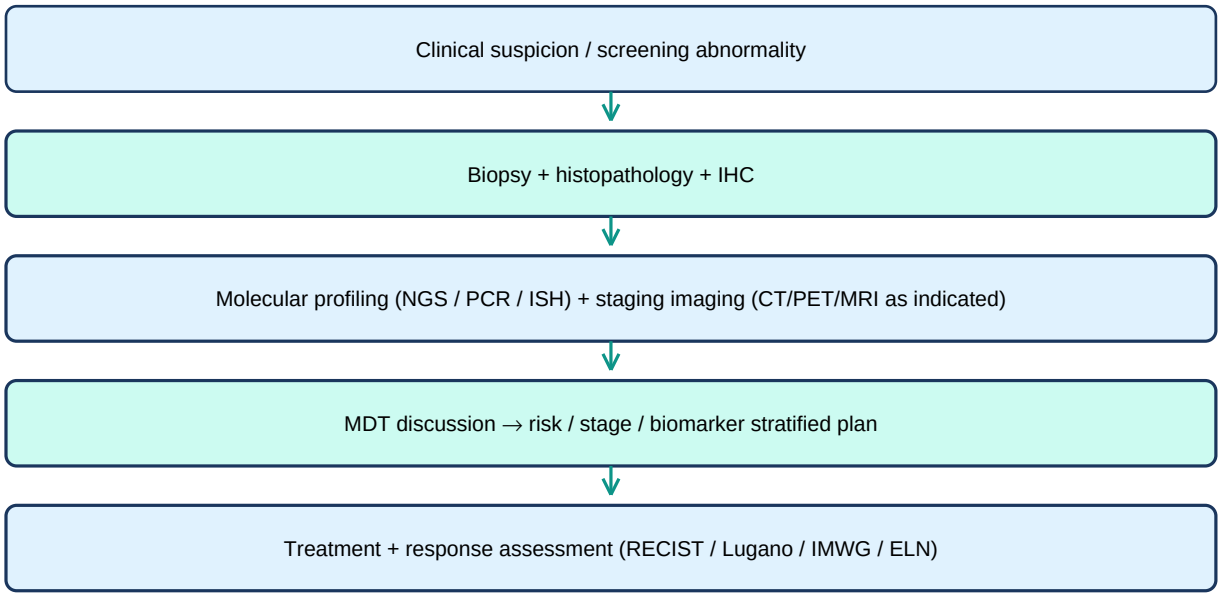


Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.
- Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).
- Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.
- Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).
- Re-biopsy / liquid biopsy at progression to map resistance.

Management flowchart (reproduce in exam)

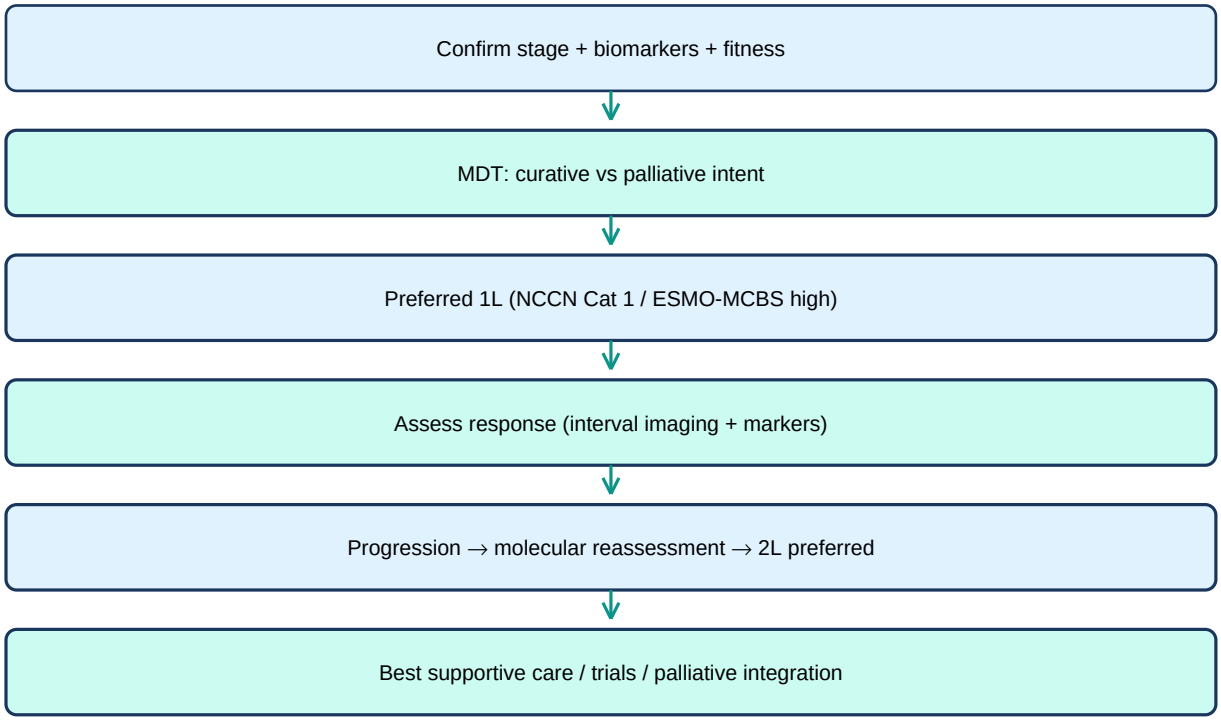


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

- Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.
- Distinguish statistical vs clinical significance; note crossover and subgroup caveats.

• For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred
ZUMA-1 / JULIET / TRANSCEND	Axi-cel; tisa-cel; liso-cel LBCL
ZUMA-7 / TRANSFORM / BELINDA	2L CAR-T vs SOC

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

• Anticipate class toxicities and counsel patients; use CTCAE grading.

• Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.

• IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.

• ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.

• TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.

• Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

Toxicity management loop

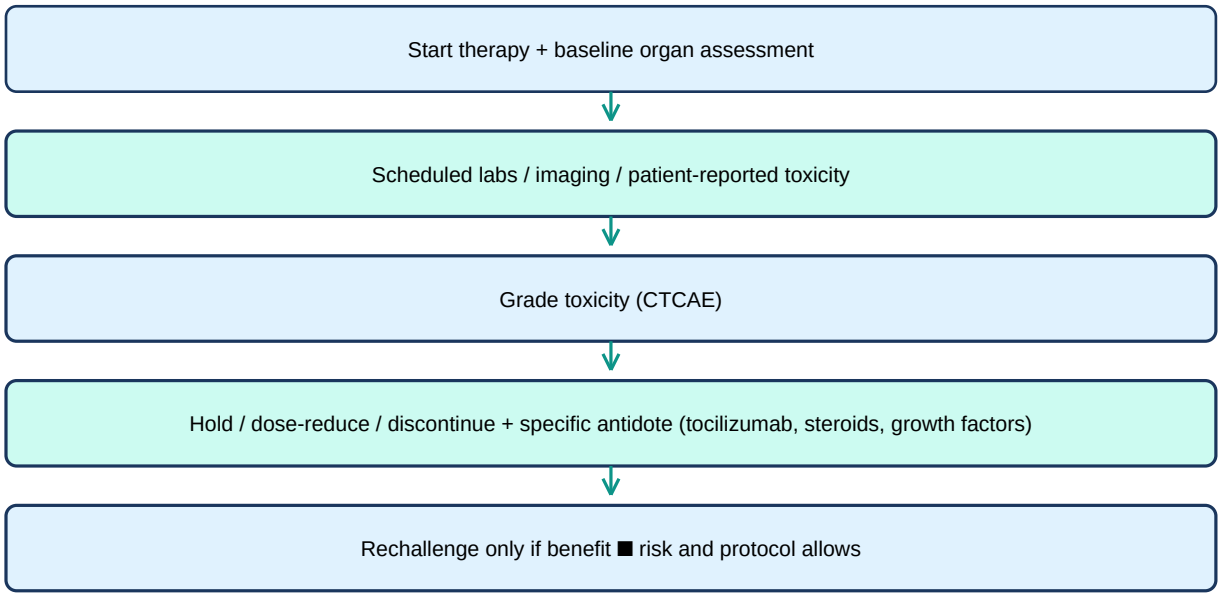


Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

• State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.

- If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* B-Cell Lymphomas v4.2026
- Earlier-line CPI expansion: ATOMIC stage III dMMR colon; neoadjuvant CPI for dMMR rectal; neoadjuvant melanoma; EV+P bladder; FLOT+durvalumab gastric.
- PD-L1 testing emphasized for gastric/EGJ; limited benefit if PD-L1 negative.

EXAM PEARLS / HALLMARKS

- NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).
- Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

Discuss differentials, mimics, and when to refer to radiation oncology, surgery, palliative care, genetics, or cell-therapy centers.

Include brief notes on health economics (cost-effectiveness, biosimilars) and clinical trial enrollment as standard options.

- MDT documentation improves outcomes and exam completeness.
- Survivorship care plans: late effects, second malignancies, cardiac, endocrine, psychosocial.
- Real-world evidence and digital tools increasingly complement RCT data (relevant to 2026 Paper IV themes).

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

- Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.
- Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.
- Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.
- Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

- Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)
- Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 15-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification

- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

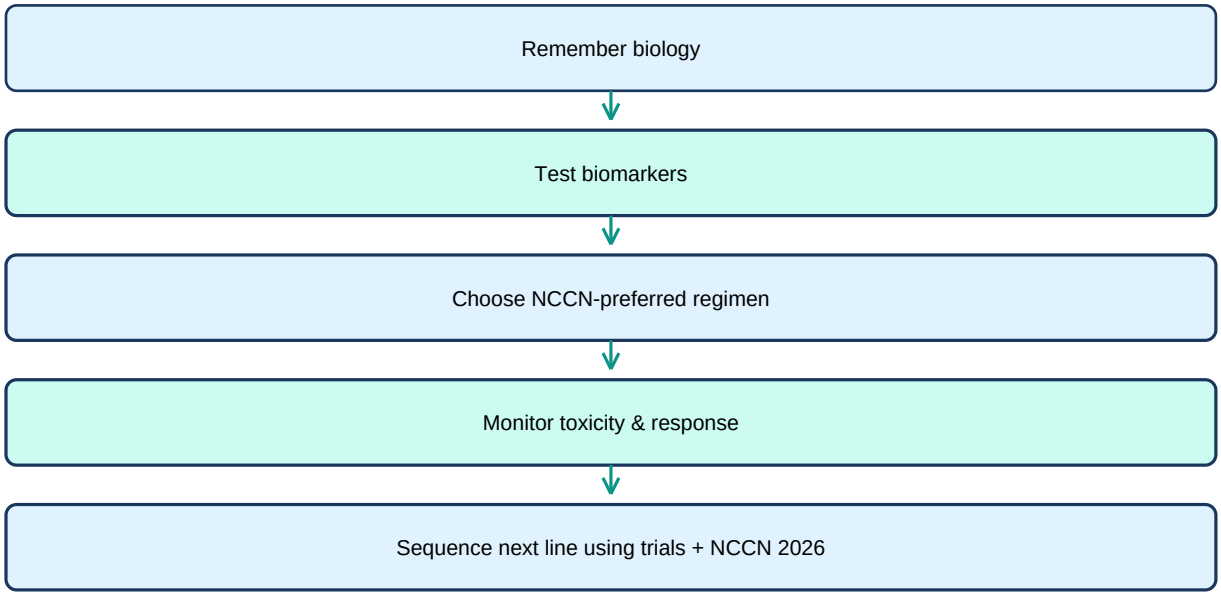


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say 'NCCN 2026' explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.
- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.
- ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

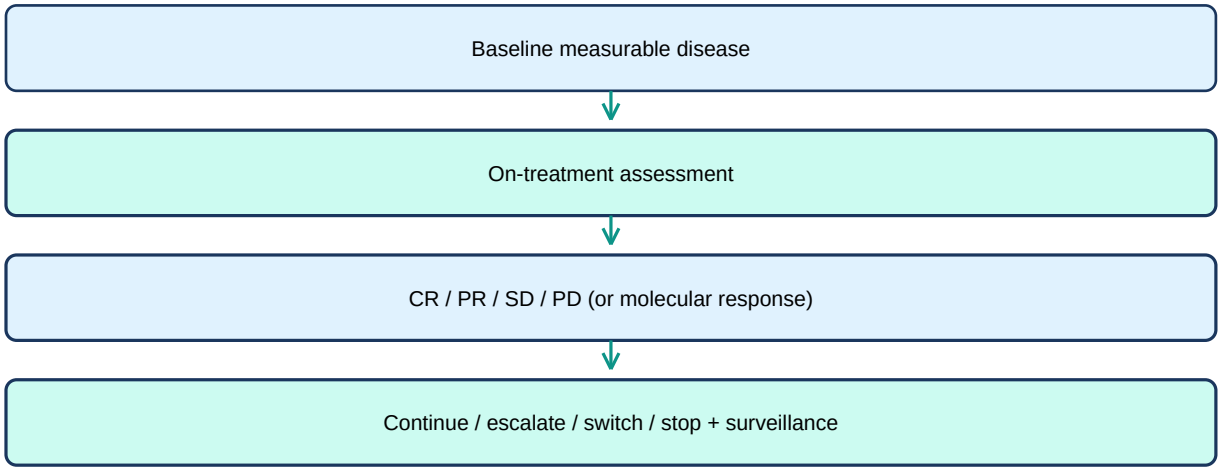


Figure: stepwise exam algorithm

13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

- Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.
- Another table: resistance mutation → preferred next drug.

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred

Table: landmark clinical trials

14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

- Document MDT decisions; clinical trial referral when standard options exhausted.
- Counsel financial toxicity transparently; prioritize highest-value Category 1 options.
- Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

15. Sample ‘examiner hot spots’ & model closing paragraph

Model closing: “In summary, management of this problem is biomarker-driven. Current standards are supported by phase III trials and reflected in NCCN 2026/2027 updates. Sequencing should maximize OS and QoL while mitigating class-specific toxicities. Enrollment in clinical trials and adherence to MDT pathways remain integral.”

Hot spots: exact doses, one HR/median, resistance mechanism, NCCN category, and a drawn algorithm.

EXAM PEARLS / HALLMARKS

- If stuck on memory: write mechanism → class → one trial → one dose → one toxicity → NCCN 2026 line.
- Never contradict yourself between algorithm and trial table.

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

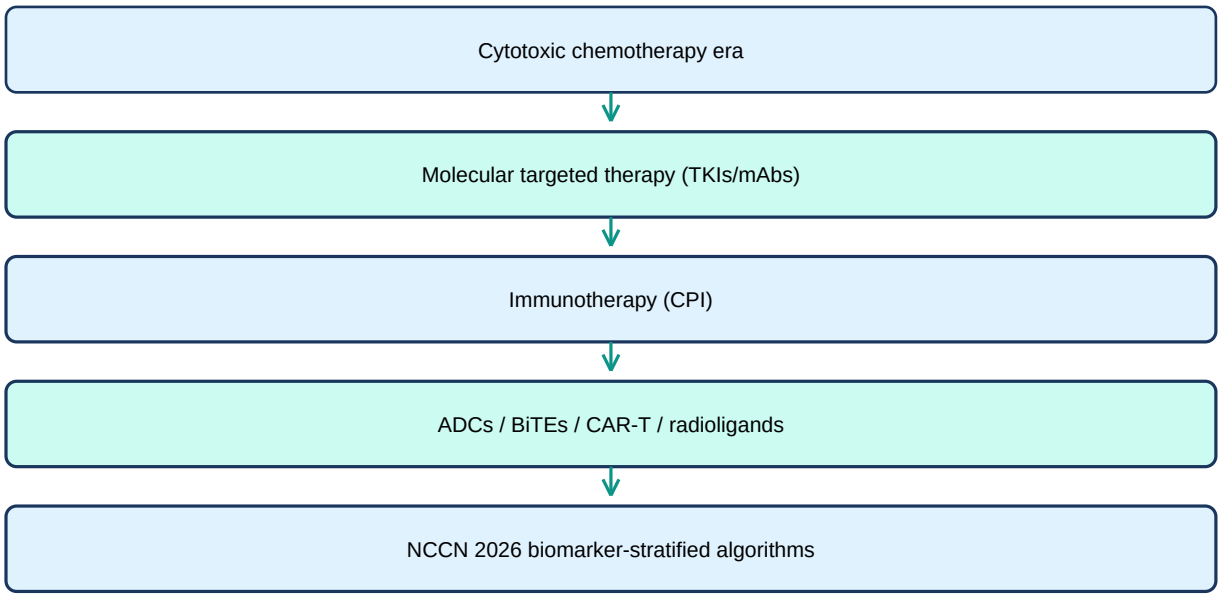


Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

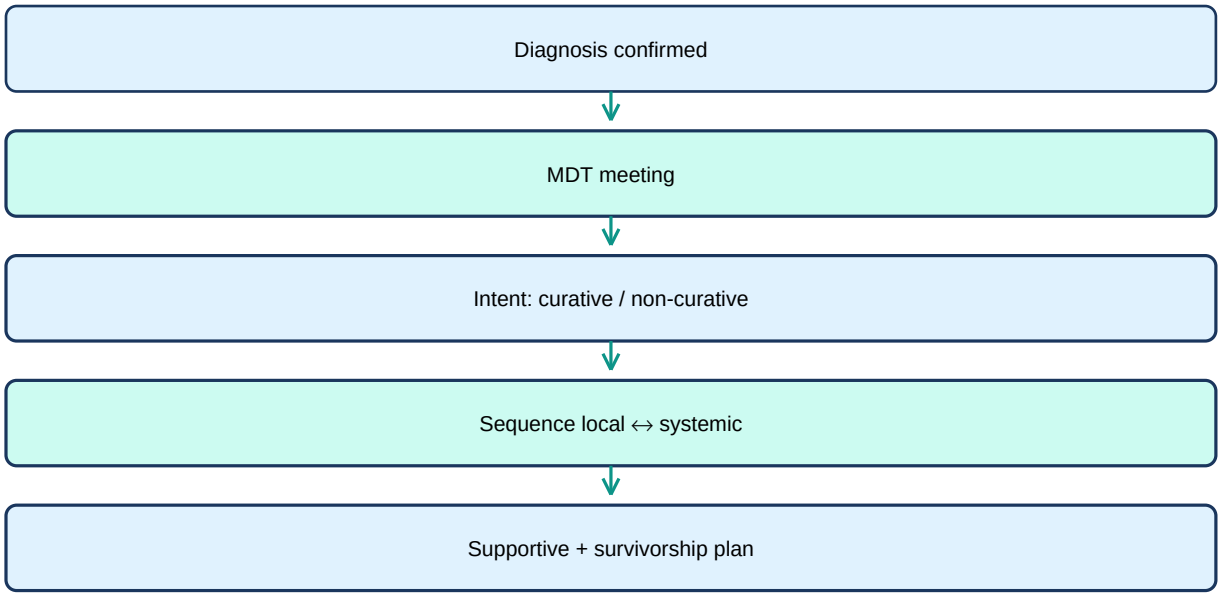


Figure: stepwise exam algorithm

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.

• NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

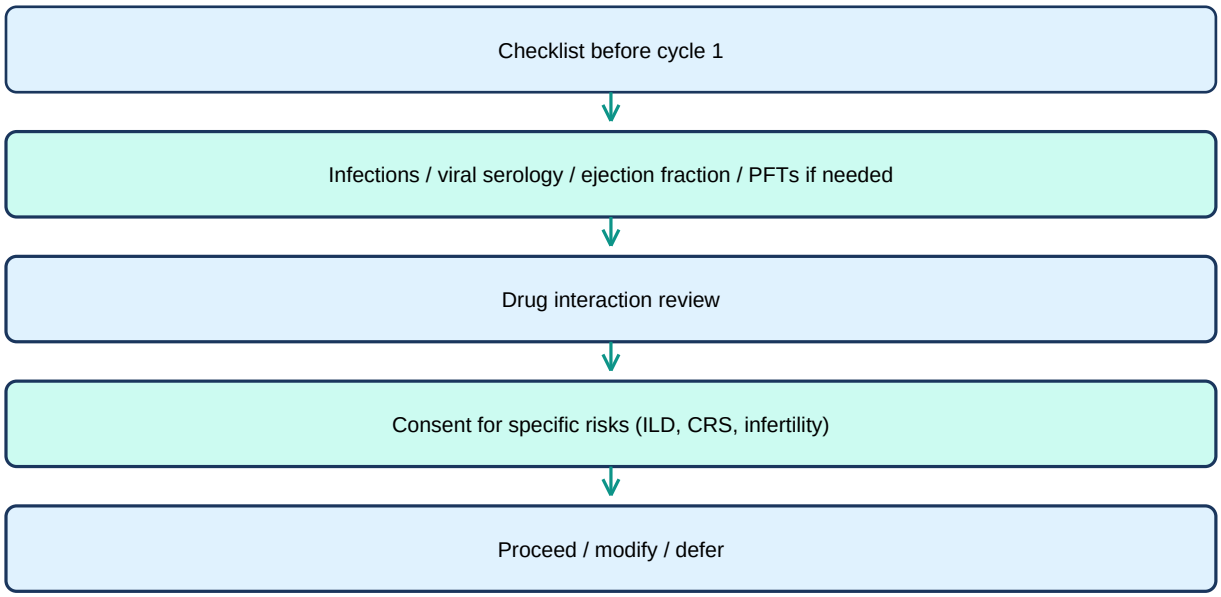


Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.

- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

- Wall chart + flowchart + trial table = visual completeness.
- Write ‘NCCN 2026’ in quadrant 4 explicitly.

I2. Full-length essay addendum — tables to copy into the answer book

For 15-mark elaborations, add two hand-drawn tables: (Table A) Drug–dose–pearl; (Table B) Trial–population–endpoint–practice change. Then write a short paragraph on quality-of-life and patient-reported outcomes, which modern trials increasingly report.

Discuss biomarker evolution on therapy (liquid biopsy cadence), and when to stop therapy (duration of CPI, TFR in CML, maintenance stopping rules).

Include a brief note on second malignancies and late effects when curative-intent therapy is discussed (alkylators, RT fields, PARPi MDS/AML risk, CHL survivors).

- Table A columns: Drug | Dose | Key toxicity | Pearl
- Table B columns: Trial | Arm | Primary endpoint | Result | Change in practice
- One paragraph on shared decision-making and financial counseling

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred
ZUMA-1 / JULIET / TRANSCEND	Axi-cel; tisa-cel; liso-cel LBCL
ZUMA-7 / TRANSFORM / BELINDA	2L CAR-T vs SOC

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W

Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* B-Cell Lymphomas v4.2026

• Earlier-line CPI expansion: ATOMIC stage III dMMR colon; neoadjuvant CPI for dMMR rectal; neoadjuvant melanoma; EV+P bladder; FLOT+durvalumab gastric.

• PD-L1 testing emphasized for gastric/EGJ; limited benefit if PD-L1 negative.

I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one ‘if progression then…’ resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway
- Flash 3: Mandatory biomarker
- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn
- Flash 8: 2026 guideline change
- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

EXAM PEARLS / HALLMARKS

• If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.

• Keep handwriting large for doses and HRs.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* B-Cell Lymphomas v4.2026

• Earlier-line CPI expansion: ATOMIC stage III dMMR colon; neoadjuvant CPI for dMMR rectal; neoadjuvant melanoma; EV+P bladder; FLOT+durvalumab gastric.

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Final NCCN 2026 box

Copy these bullets onto the last page of your answer booklet for Paper IV currency marks.

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* B-Cell Lymphomas v4.2026

• Earlier-line CPI expansion: ATOMIC stage III dMMR colon; neoadjuvant CPI for dMMR rectal; neoadjuvant melanoma; EV+P bladder; FLOT+durvalumab gastric.

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Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.